GEOMETRIC REDUCEDNESS FROM GENERIC POINTS: SPECIALISATION, THE REDUCED LOCAL RING CRITERION AND THE FUNCTION FIELD CASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Part (3) is dismissed by the author with the single sentence 'Part (3) follows trivially from part (2)'; that sentence is reproduced verbatim and no argument is missing from the source.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-generic-points-geometrically-reduced is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Pointwise criterion for geometric reducedness of a scheme at a point. Same chapter, Lemma lemma-geometrically-reduced-at-point, invoked at proof lines 576-577 and 589-590. Supplies the translation between the scheme statement and statements about local rings.
- Geometric reducedness is preserved by localization. Algebra, Lemma algebra-lemma-geometrically-reduced-permanence, invoked at proof lines 579-581. Supplies part (1).
- A reduced ring whose localizations at minimal primes are geometrically reduced is geometrically reduced. Algebra, Lemma algebra-lemma-generic-points-geometrically-reduced, invoked at proof lines 587-588. Supplies the main step of part (2).
- Characterization of separable field extensions. Algebra, Lemma algebra-lemma-characterize-separable-field-extensions, invoked at proof lines 594-595. Supplies part (4) from part (3).

Source reference. Original lemma 030W: sources/stacks-chow/tags.txt, source lines 3963–3963. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 07K2: sources/stacks-chow/tags.txt, source lines 9642–9642. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04KN: sources/stacks-chow/tags.txt, source lines 5979–5979. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 035W: sources/stacks-varieties/pinned-original-source.tex, source lines 354–354. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k.*

(1) *If $x' \rightsquigarrow x$ is a specialization and X is geometrically reduced at x, then X is geometrically reduced at x'.*
(2) *If $x \in X$ such that (a) $\mathcal{O}_{X,x}$ is reduced, and (b) for each specialization $x' \rightsquigarrow x$ where x' is a generic point of an irreducible component of X the scheme X is geometrically reduced at x', then X is geometrically reduced at x.*
(3) *If X is reduced and geometrically reduced at all generic points of irreducible components of X, then X is geometrically reduced.*
(4) *If X is a variety over k, then X is geometrically reduced if and only if its function field is a separable extension of k.*

Proof. Part (1) follows from Lemma 035W and the fact that if A is a geometrically reduced k-algebra, then $S^{-1}A$ is a geometrically reduced k-algebra for any multiplicative subset S of A, see Algebra, Lemma 04KN.

Let $A = \mathcal{O}_{X,x}$. The assumptions (a) and (b) of (2) imply that A is reduced, and that $A_{\mathfrak{q}}$ is geometrically reduced over k for every minimal prime $\mathfrak{q}$ of A. Hence A is geometrically reduced over k, see Algebra, Lemma 07K2. Thus X is geometrically reduced at x, see Lemma 035W.

Part (3) follows trivially from part (2). Finally, (4) follows from (3) by Algebra, Lemma 030W. □